

Project write-up notes

Research Plan & Experimental Roadmap

May 12, 2026

1 Framing the project

This project is best situated as *lifted generalized-policy synthesis by grammar-guided MCTS* — not as full AlphaZero, at least not yet. The search machinery is Monte Carlo Tree Search, but it is steered by a grammar over candidate policies rather than by a value/policy network learned from self-play over a fixed game; and the object of synthesis is a *lifted* (relational, object-count-agnostic) generalized policy rather than a tabular or instance-specific one.

[milton: “not full AlphaZero yet” — worth spelling out which AlphaZero ingredients are deliberately deferred (a learned value net? the self-play loop? policy iteration?), so the contrast is concrete.]

2 Generalized planning, briefly

Generalized planning concerns policies or plans that solve *multiple* instances of a domain — including the cases where the set of objects and the set of ground actions change from one instance to the next. This is exactly the regime a lifted policy is built for: a solution expressed over predicates and quantified variables, rather than over a frozen ground state space, transfers across instances with different object sets.

Bonet and Geffner explicitly motivate generalized policies for relational domains where the object set varies. [milton: add the actual citation key once the reference is in the .bib — the input named “Bonet and Geffner” but not a specific paper, so no \cite{} is inserted yet.]